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(54) **OPTICAL INSPECTION SYSTEM, METHOD OF OPTICAL INSPECTION, AND METHOD OF MANUFACTURING SEMICONDUCTOR PACKAGE**

(71) Applicant: **Taiwan Semiconductor Manufacturing Company, Ltd.**, Hsinchu (TW)

(72) Inventors: **Zih-Ying Yang**, Hsinchu City (TW); **Yang-Chih Hsueh**, Hsinchu City (TW); **Ming-Tsu Chung**, Hsinchu (TW); **Yan-Zuo Tsai**, Hsinchu City (TW); **Yung-Chi Lin**, New Taipei City (TW)

(73) Assignee: **Taiwan Semiconductor Manufacturing Company, Ltd.**, Hsinchu (TW)

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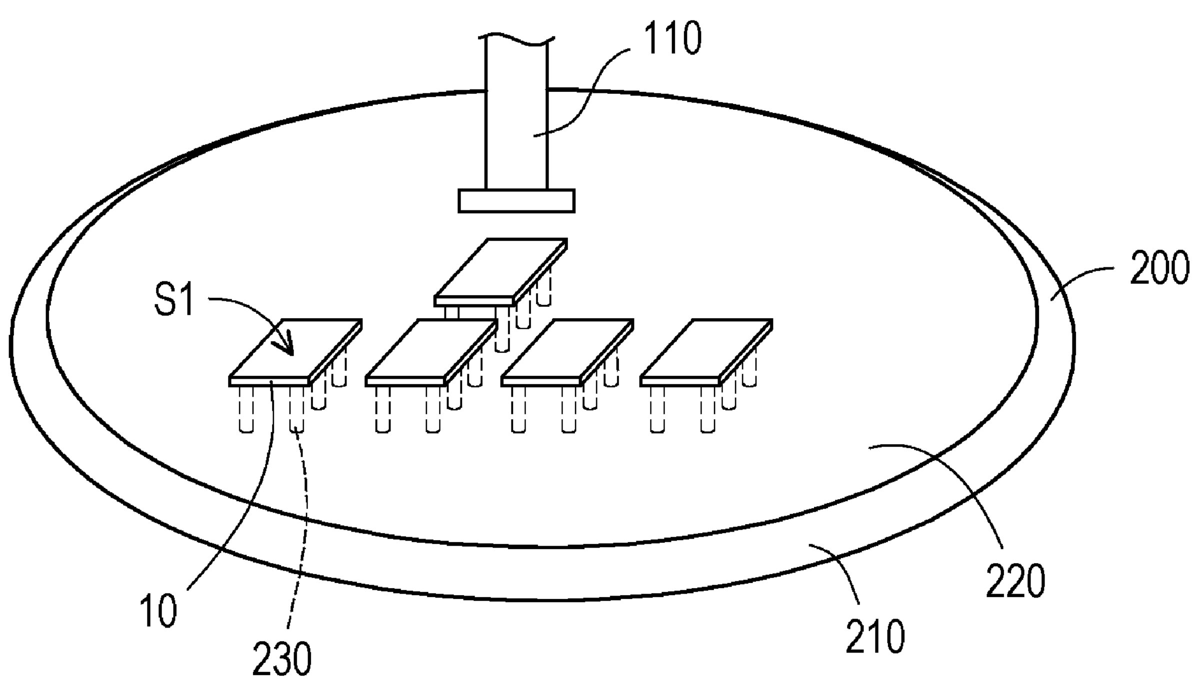
(57) **ABSTRACT**

An optical inspection system includes a die carrier, a camera device, and a processor. The die carrier is configured to carry a die, wherein a backside surface of the die faces away from the die carrier. The camera device is movably disposed above the die carrier and configured to capture images of a plurality of detecting spots on the backside surface, wherein the plurality of detecting spots comprising a plurality of die corners of the die. The processor is coupled to the camera device and configured to determine a die center of the die according to the images of the plurality of die corners of the die, and determine whether the die is a bad die according to an existence of a defect on the backside surface of the die detected by the camera device.



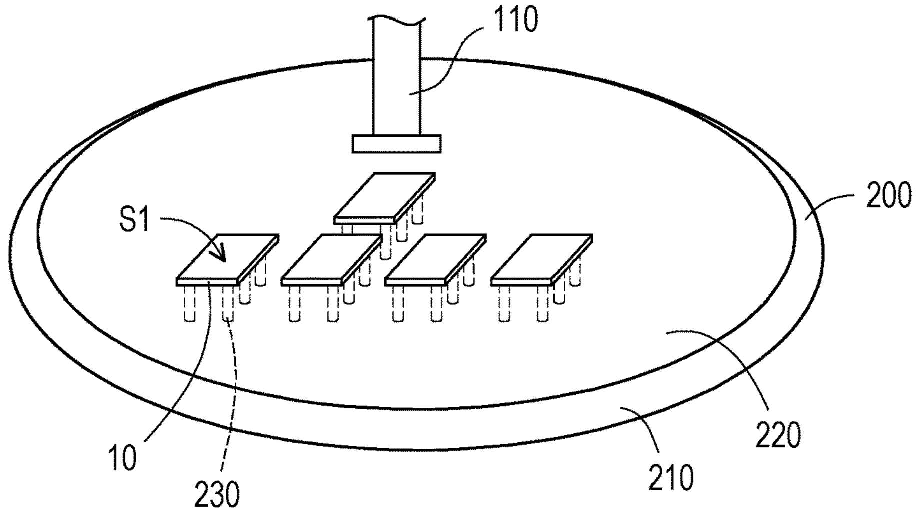


FIG. 1

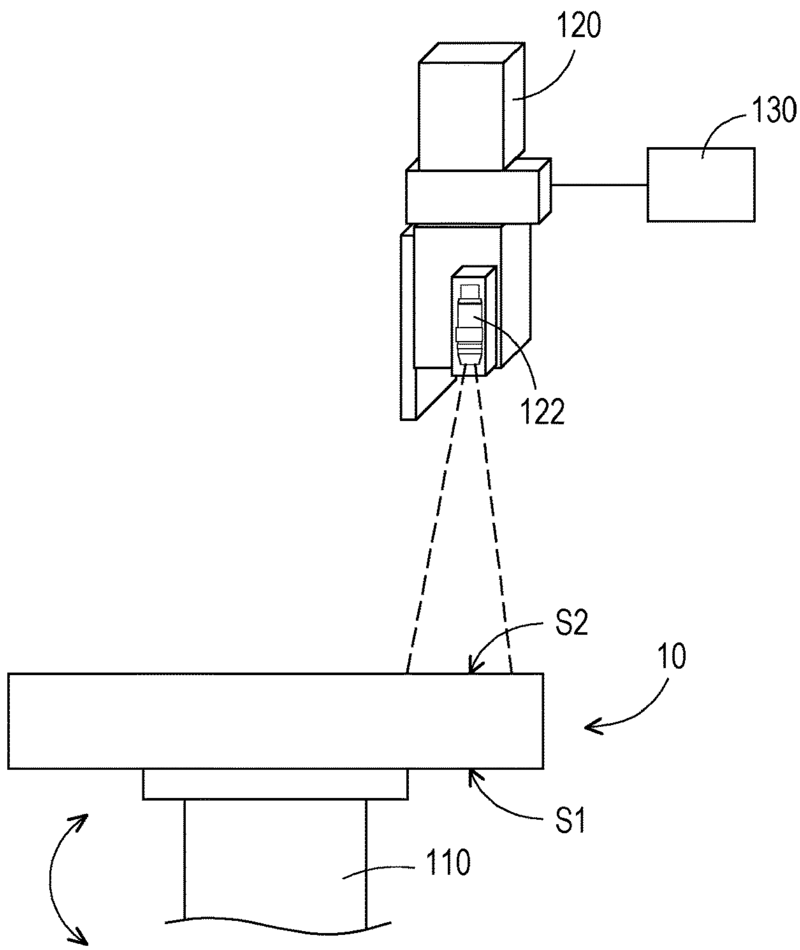


FIG. 2

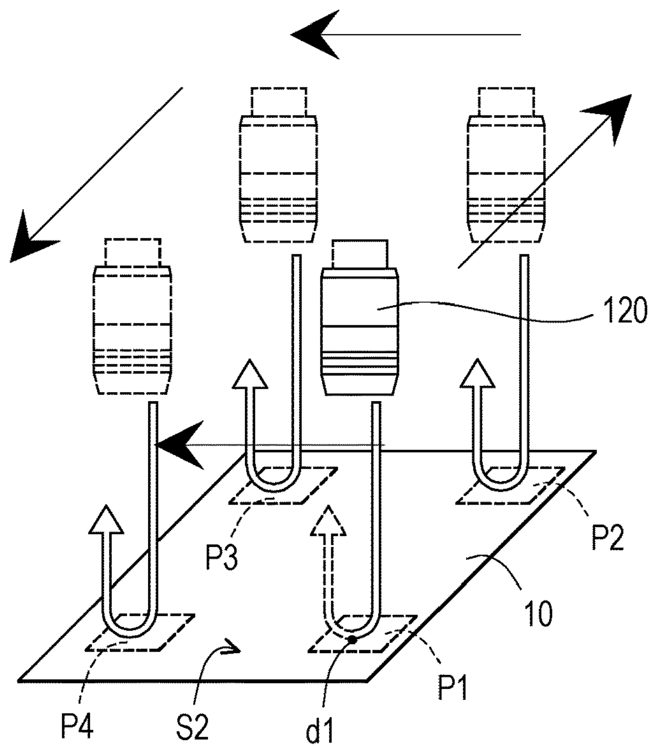


FIG. 3

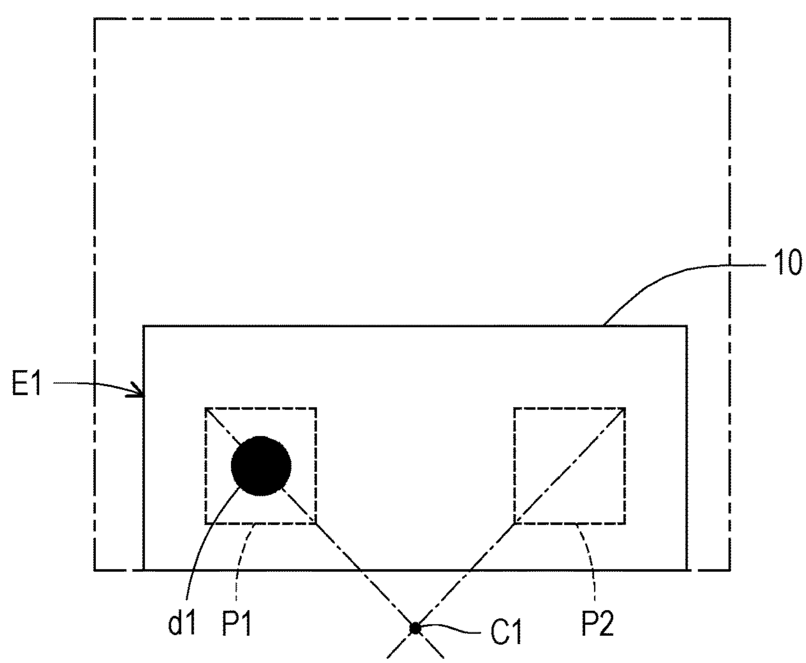


FIG. 4

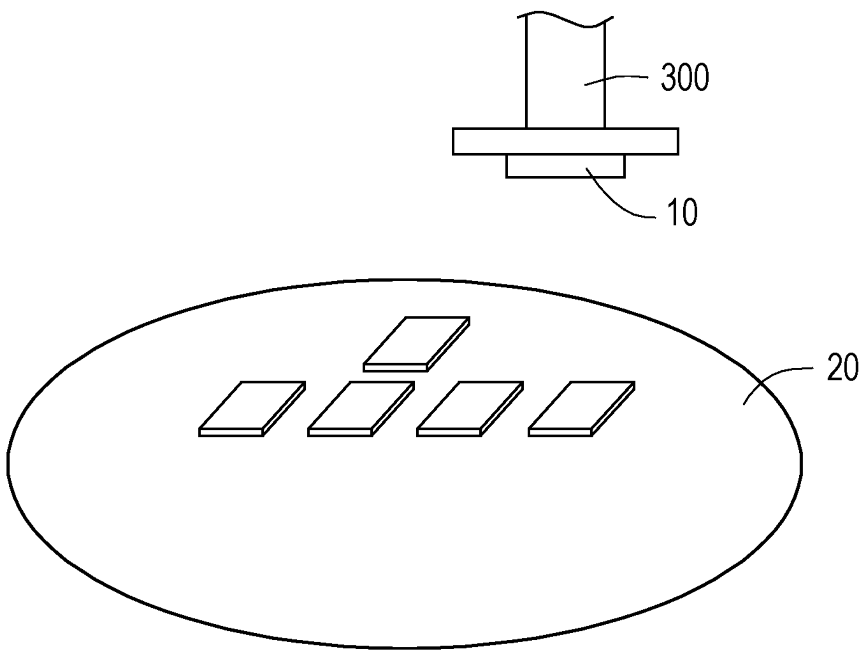


FIG. 5

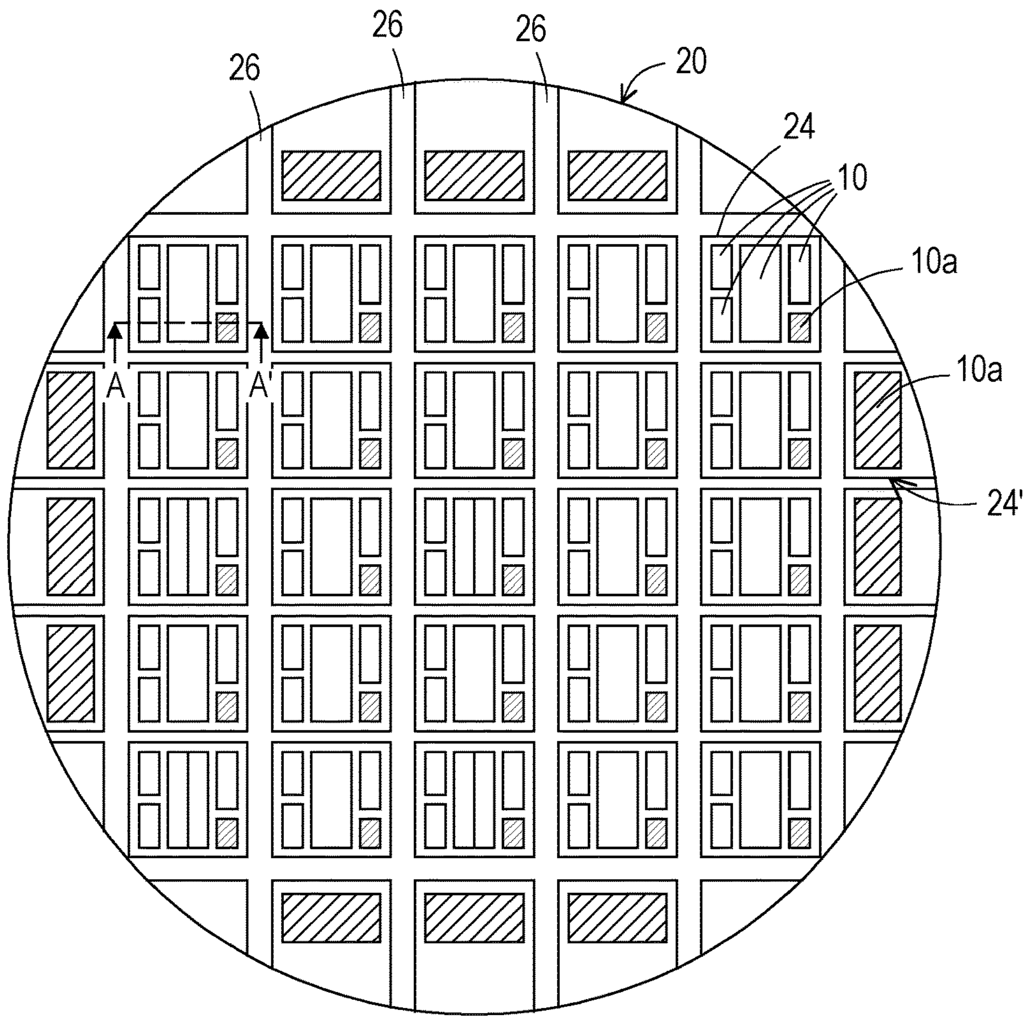

26
26
26
20
24
10
10a
10a
A
A'
24'

FIG. 6

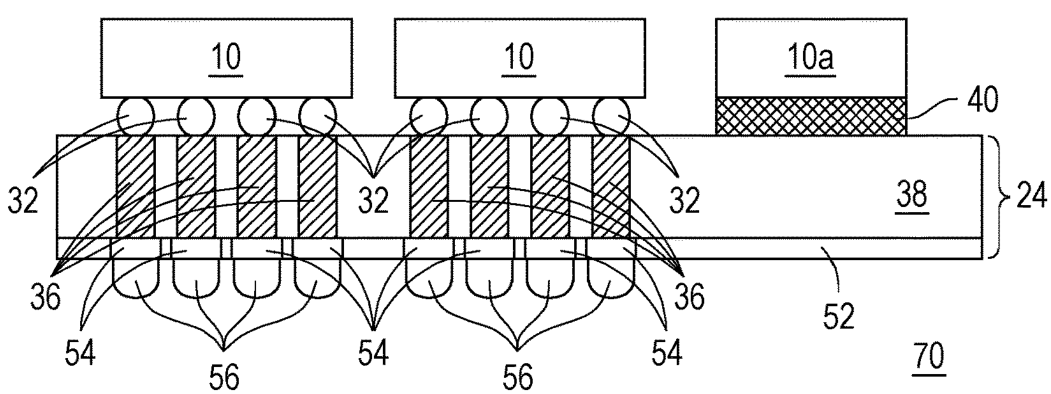

10
10
10a
40
32
32
32
38
24
36
36
52
54
54
54
56
56
70

FIG. 7

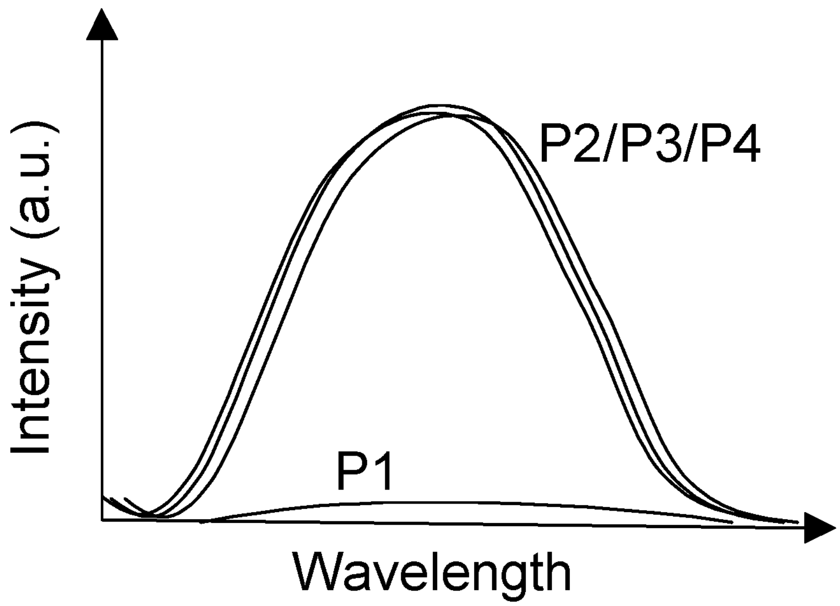


FIG. 8

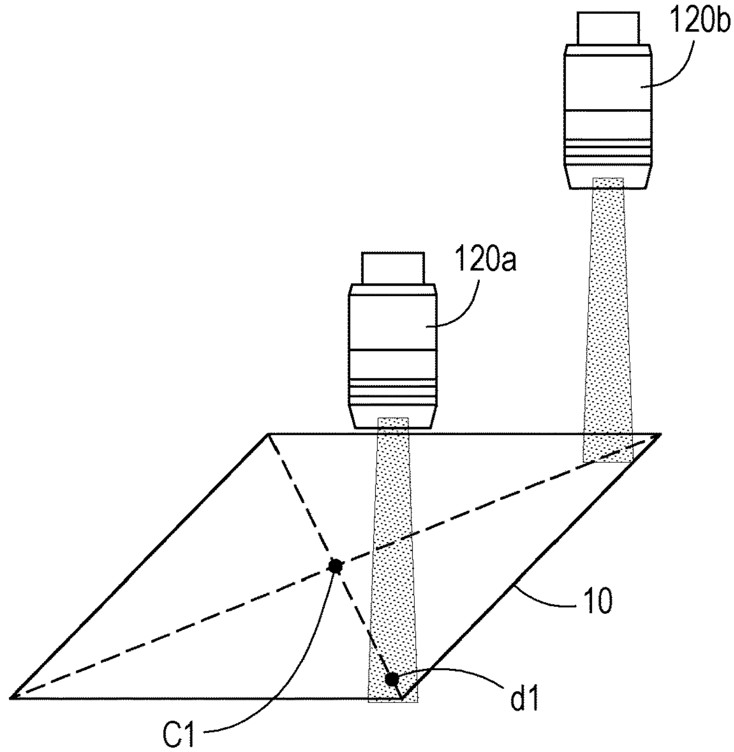


FIG. 9

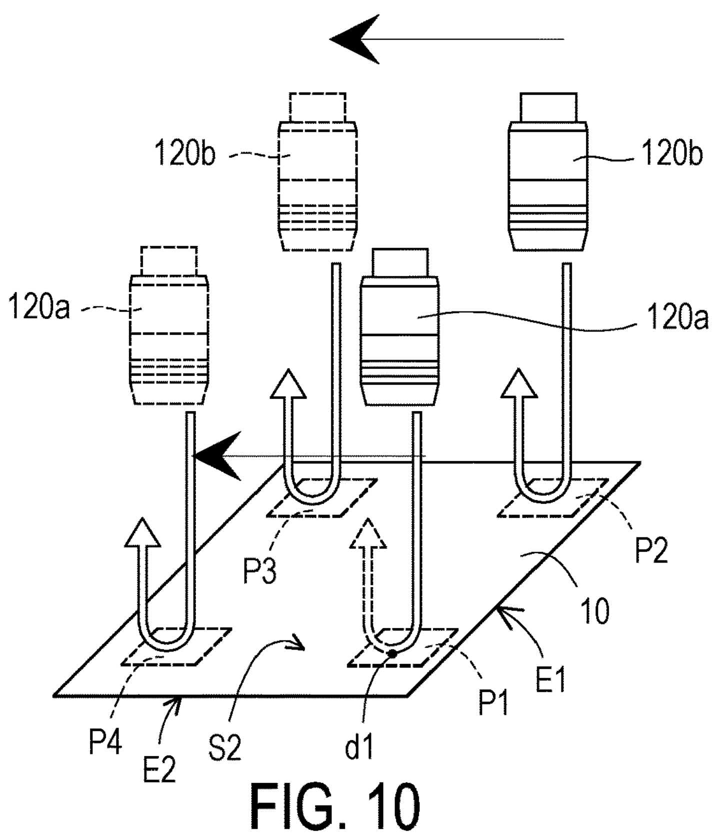


FIG. 10

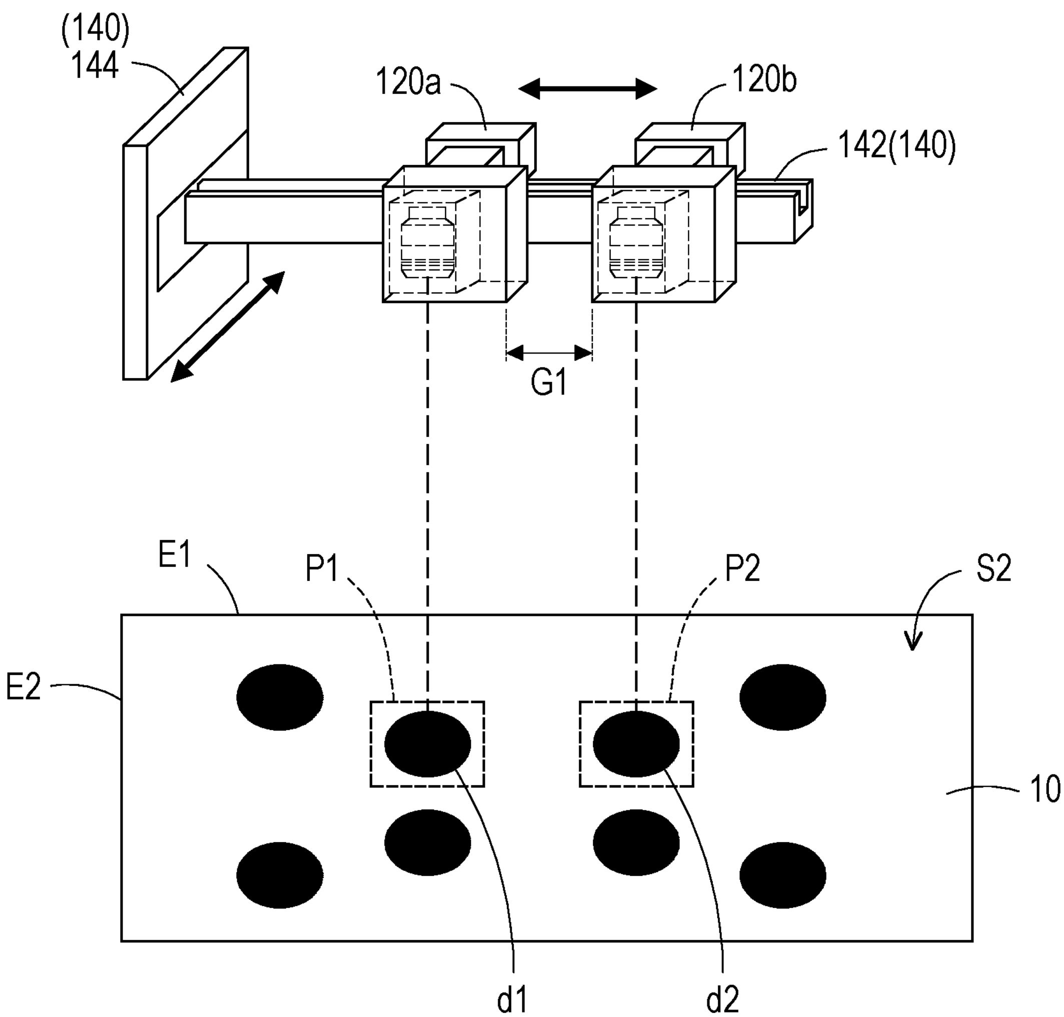


FIG. 11

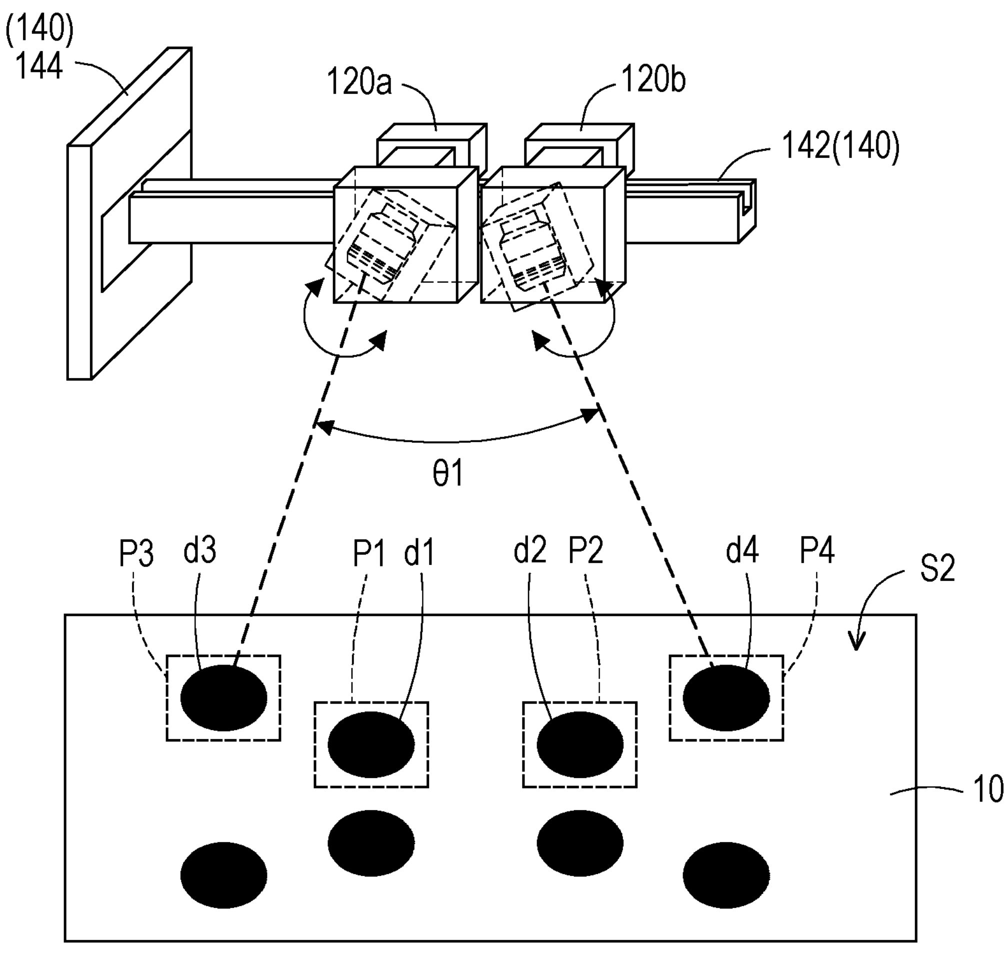


FIG. 12

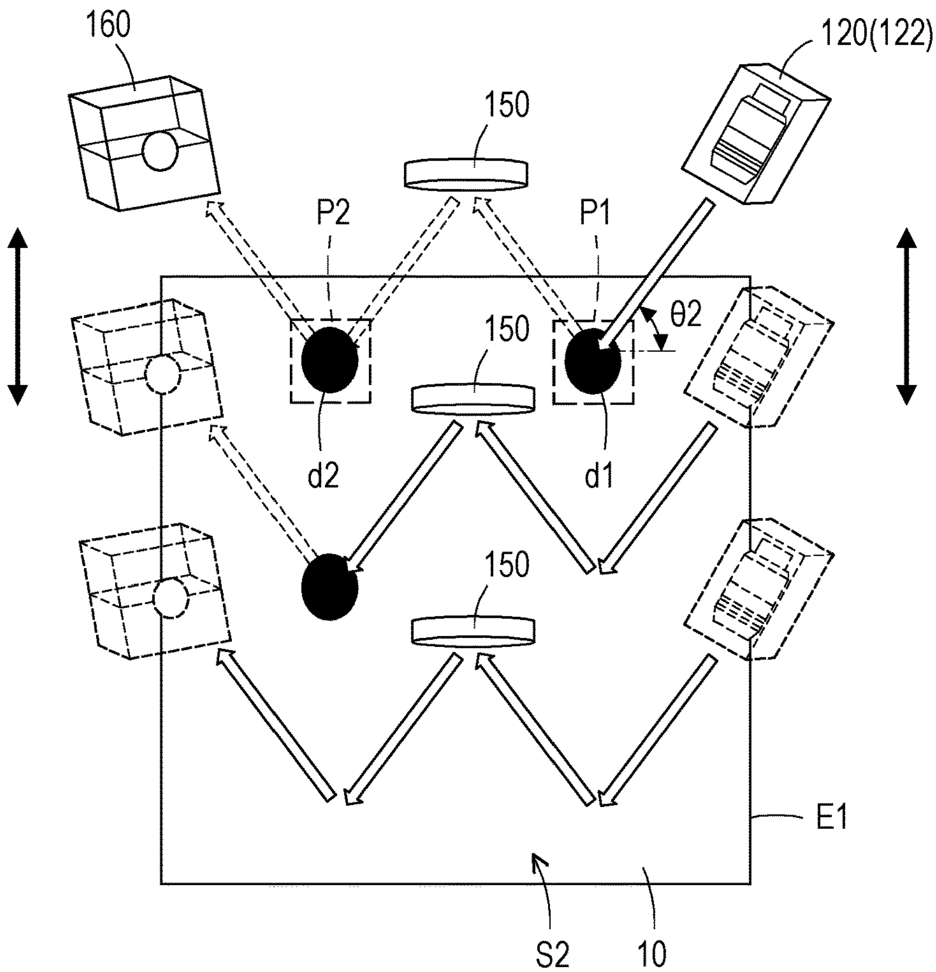


FIG. 13

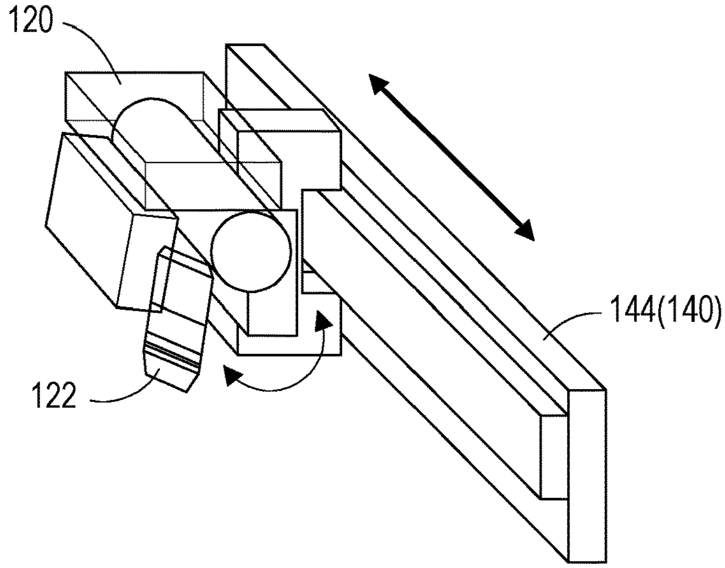


FIG. 14

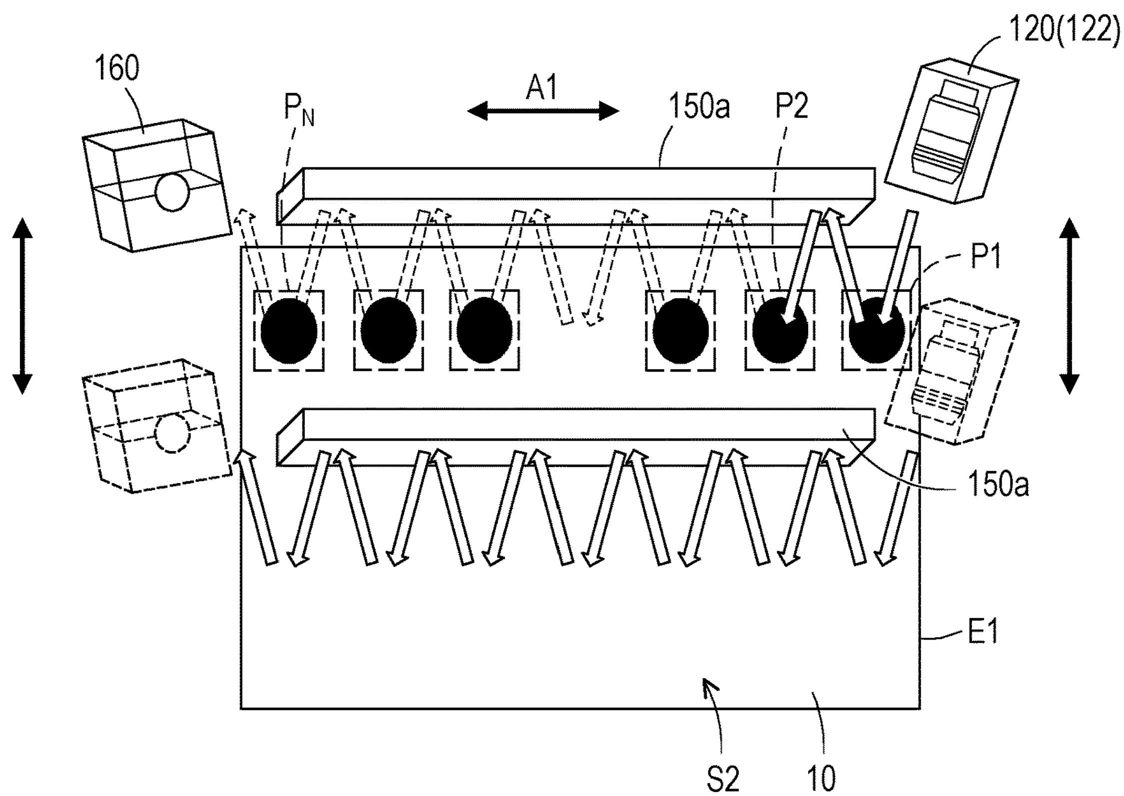

160
P_N
A1
150a
P2
120(122)
P1
150a
E1
S2
10

FIG. 15

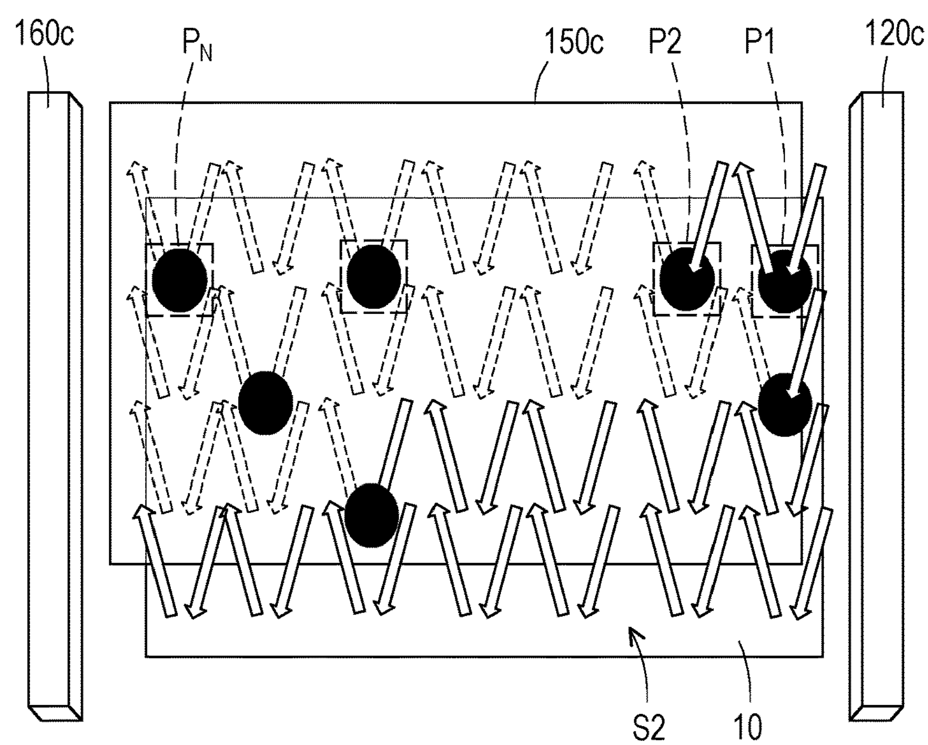

160c
P_N
150c
P2
P1
120c
S2
10

FIG. 16

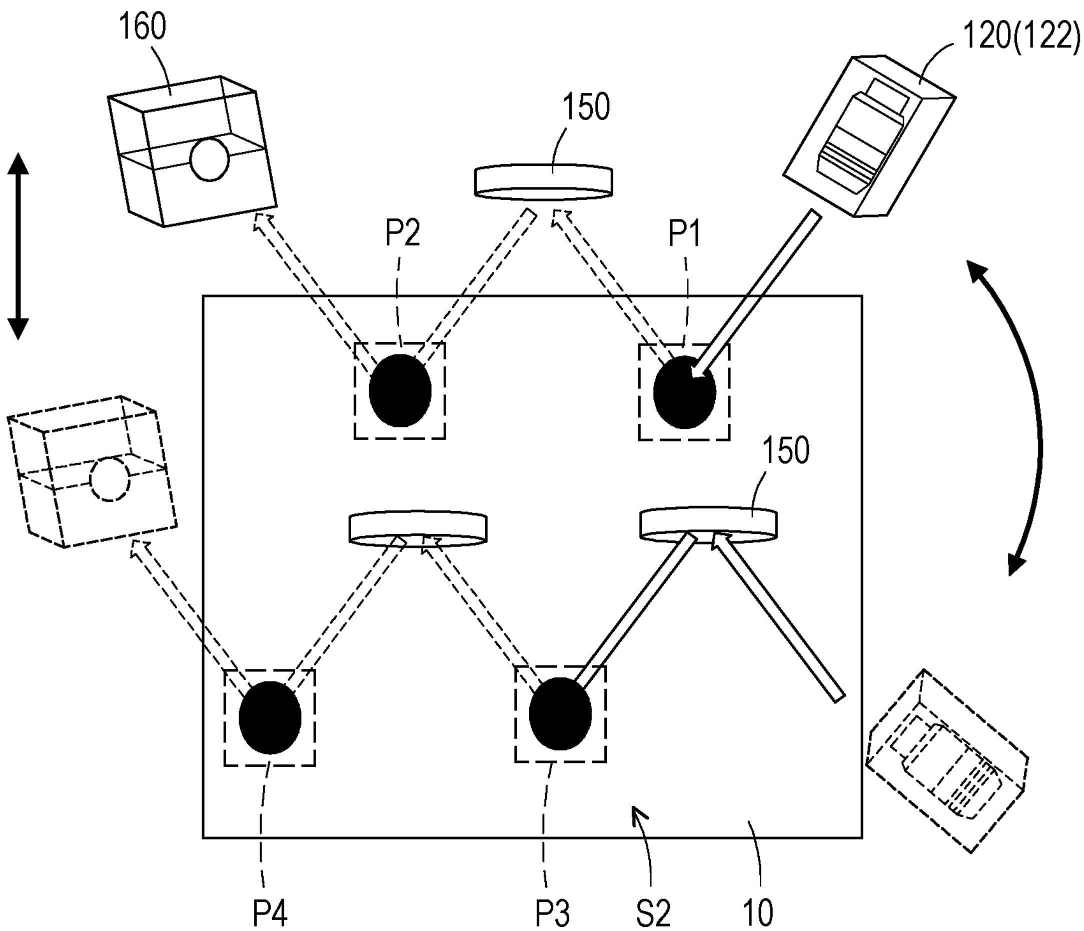


FIG. 17

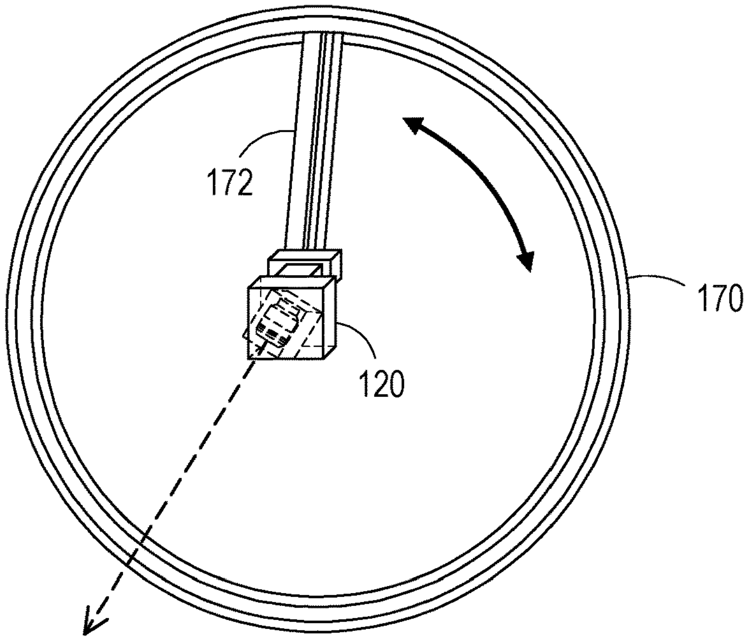


FIG. 18

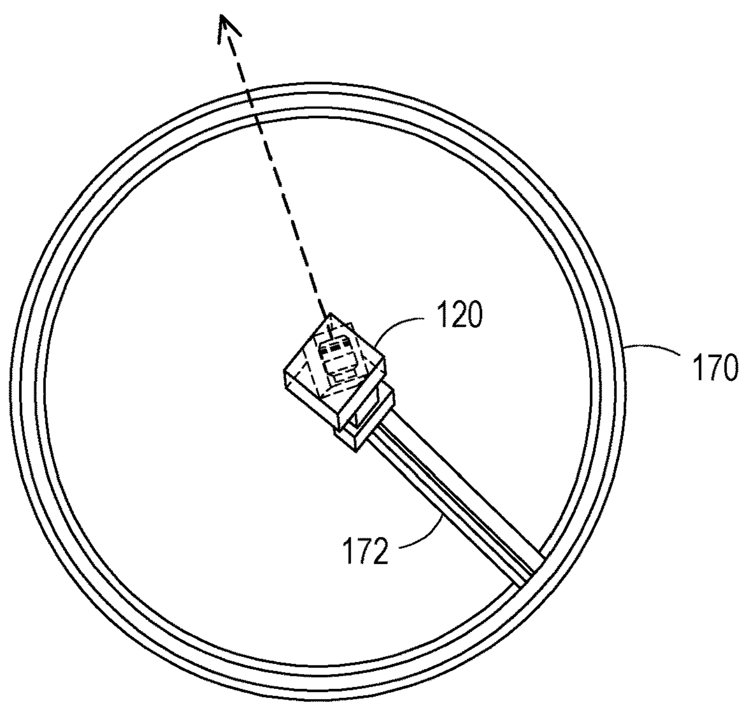


FIG. 19

OPTICAL INSPECTION SYSTEM, METHOD OF OPTICAL INSPECTION, AND METHOD OF MANUFACTURING SEMICONDUCTOR PACKAGE

BACKGROUND

[0001] The semiconductor industry has experienced rapid growth due to ongoing improvements in the integration density of a variety of electronic components (e.g., transistors, diodes, resistors, capacitors, etc.). For the most part, improvement in integration density has resulted from iterative reduction of minimum feature size, which allows more components to be integrated into a given area. As the demand for shrinking electronic devices has grown, a need for smaller and more creative packaging techniques of semiconductor dies has emerged. An example of such packaging systems is Package-on-Package (PoP) technology. In a PoP device, a top semiconductor package is stacked on top of a bottom semiconductor package to provide a high level of integration and component density. PoP technology generally enables production of semiconductor devices with enhanced functionalities and small footprints on a printed circuit board (PCB).

BRIEF DESCRIPTION OF THE DRAWINGS

[0002] Aspects of the present disclosure are best understood from the following detailed description when read with the accompanying figures. It is noted that, in accordance with the standard practice in the industry, various features are not drawn to scale. In fact, the dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion.

[0003] FIG. **1** to FIG. **7** illustrate schematic views of intermediate stages in the manufacturing of a semiconductor package according to some exemplary embodiments of the present disclosure.

[0004] FIG. **8** is a diagram illustrating a relationship between wavelength and intensity of light according to some exemplary embodiments of the present disclosure.

[0005] FIG. **9** and FIG. **10** illustrate schematic views of camera devices capturing images of a plurality of detecting spots on a die according to some exemplary embodiments of the present disclosure.

[0006] FIG. **11** and FIG. **12** illustrate schematic views of camera devices capturing images of a plurality of detecting spots on a die according to some exemplary embodiments of the present disclosure.

[0007] FIG. **13** illustrates a schematic view of a camera device capturing images of a plurality of detecting spots on a die according to some exemplary embodiments of the present disclosure.

[0008] FIG. **14** illustrates a schematic view of a camera device disposed on a moving mechanism according to some exemplary embodiments of the present disclosure.

[0009] FIG. **15** illustrates a schematic view of a camera device capturing images of a plurality of detecting spots on a die according to some exemplary embodiments of the present disclosure.

[0010] FIG. **16** illustrates a schematic view of a line-scan camera device capturing images of a plurality of detecting spots on a die according to some exemplary embodiments of the present disclosure.

[0011] FIG. **17** illustrates a schematic view of a camera device disposed on a moving mechanism according to some exemplary embodiments of the present disclosure.

[0012] FIG. **18** and FIG. **19** illustrate schematic views of a camera device disposed on a moving mechanism according to some exemplary embodiments of the present disclosure.

DETAILED DESCRIPTION

[0013] The following disclosure provides many different embodiments, or examples, for implementing different features of the provided subject matter. Specific examples of components and arrangements are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. For example, the formation of a first feature over or on a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact, and may also include embodiments in which additional features may be formed between the first and second features, such that the first and second features may not be in direct contact. In addition, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments and/or configurations discussed.

[0014] Further, spatially relative terms, such as "beneath," "below," "lower," "above," "upper" and the like, may be used herein for ease of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly.

[0015] The optical inspection system and methods provided in the disclosure are used to evaluate to what extent defects are present on a semiconductor die. For example, the optical inspection system can be used to optically inspect and detect tape residue, and/or particle contaminants on the surface of the die before the die is bonded to another workpiece such as a die, an interposer, a substrate, a wafer, or the like. Accordingly, some aspects of the present disclosure provide for optical inspection system and methods by which dies can be inspected in-situ with limited, if any, human handling/transport. As such, these optical inspection system and methods limit labor costs and also minimize breakage, contamination, and defects for dies and improve yields of subsequent bonding process.

[0016] FIG. **1** to FIG. **7** illustrate schematic views of intermediate stages in the manufacturing of a semiconductor package according to some exemplary embodiments of the present disclosure. A method of manufacturing a semiconductor package may include the following steps. In some embodiments, a semiconductor wafer is provided on a tape carrier **200**. For example, the tape carrier **200** includes an adhesive tape **220**, which is supported and fixed to a ring flame **220**. The semiconductor wafer may be press-bonded onto the adhesive tape **220** of the tape carrier **200**. Referring to FIG. **1**, the dicing process is performed to a plurality of scribe lines on the semiconductor wafer to form a plurality of dies **10** on the tape carrier **200**. In some embodiments, the dicing process can adopt, for example, a cutting method called full-cut that forms a slit into the adhesive tape **220** of the tape carrier **200**. The dicing apparatus used in the dicing process is not particularly limited, and a conventionally known apparatus can be used. Further, since the semiconductor wafer is adhered and fixed by the tape carrier **200**, die crack and die fly can be suppressed, as well as the damage of the semiconductor wafer can also be suppressed. Consequently, the semiconductor wafer is cut into a prescribed size and individualized (is formed into small pieces) to from individual dies **10**, i.e., semiconductor chips. In some embodiments, each of the dies **10** includes a front side surface S**1**, which circuits or the like of the die **10** is formed, and a backside surface S**2** opposite to front side surface S**1**. In some embodiments, the backside surface S**2** of the die **10** is attached to the tape carrier **200** while the front side surface S**1** faces way from the tape carrier **200**.

[0017] Then, picking-up one of the plurality of dies **10** from the tape carrier **200** is performed in order to collect a dies **10** that is adhered and fixed to the tape carrier **200** for the subsequent bonding process. For example, a pick and place tool may be used to separate the individual dies **10** from the tape carrier **200**. The pick and place tool includes a die carrier **110**, which is configured to pick up an individual die **10** from the tape carrier **200**, carry the die **10**, and place the die **10** onto a substrate such as an interposer, a package substrate, a wafer or other dies. In one embodiment, the die carrier **110** may be a vacuum collet configured to lift the die **10** from the tape carrier **200** and places it onto a substrate to be bonded. In the embodiment, the die carrier **110** picks up the die **10** with its front side surface S**1** facing the die carrier **110**, and the backside surface S**2** faces away from the die carrier **110**. In some embodiments, a plurality of ejector pins **230** are disposed underneath the tape carrier **200**, and the die **10** is pushed up by the ejector pins **230**, and is picked up by vacuum with the die carrier **110**. In other words, the ejector pins **230** underneath the tape carrier **200** are ejected for pushing the die **10** away from the tape carrier **200**.

[0018] Then, referring to both FIGS. **1** and **2**, after the die **10** is picked up by the die carrier **110**, the die **110** may be flipped by the die carrier **110**, so that the backside surface S**2** of the die **10** faces up toward a camera device **120**, which is movably disposed above the die carrier **110**. The camera device **120** is configured to capture images of a plurality of detecting spots on the backside surface S**2** to find a die center and detect defects on the backside surface S**2** of the die **10**. Generally, when the die **10** is picked up from the tape carrier **200**, a part of the tape residue from the tape carrier **200** may be left on the backside surface S**2** of the die during the pickup process, especially at spots where the ejector pins **230** (shown in FIG. **1**) pushes the backside surface S**2** of the die **10**. The tape residue may lead to an uneven contact surface (i.e., backside surface S**2** with tape residue left thereon) pressed by a bonding head in the subsequent bonding process and causes poor bonding quality or even die crack. In addition, a part of the tape residue may even be stuck to the bonding head and then contaminates other dies. Accordingly, the disclosure provides system and method to detect defects on the backside surface S**2** of the die **10** before the bonding process is performed. The defects may include tape residue, particles, contaminants, or the like, left on the backside surface S**2** of the die **10**.

[0019] Referring to FIG. **2** and FIG. **3**, in accordance with some embodiments of the disclosure, the optical inspection system include the die carrier **110**, at least one camera device **120**, and a processor **130** coupled to the camera device **120**. The camera device **120** is configured to capture images of a plurality of detecting spots P**1**, P**2**, P**3**, and P**4** on the backside surface S**2** of the die **110**. It is noted that 4 detecting spots P**1**, P**2**, P**3**, and P**4** are illustrated in the present embodiment, but more or less detecting spots may also be applied, and the disclosure is not limited thereto. In the embodiment, one camera device **120** is adopted to capture images of the detecting spots P**1**, P**2**, P**3**, and P**4**, but in other embodiments, more camera devices may also be applied. In the present embodiment, the detecting spots P**1**, P**2**, P**3**, and P**4** includes a plurality of die corners of the die **10**. In some embodiments, the detecting spots may correspond to the spots where the ejector pins (e.g., the ejector pins **230** shown in FIG. **2**) pushes backside surface S**2** of the die **110**, which is the spots that tape residue is most likely to occur.

[0020] Referring to FIG. **3** and FIG. **4**, in the present embodiment, the camera device **120** can be mounted on a robot arm, so as to be moved above the die **10** (e.g., moved along arrows shown in FIG. **3**) for capturing the images of the detecting spots P**1**, P**2**, P**3**, and P**4**, such as images of the die corners. Accordingly, the processor **130** is configured to identify the borders, edges, or boundaries of the die **10** according to the images of the die corners captured by the camera device **120**. Then, the processor **130** can determine a die center C**1** of the die **10** according to the identified die edges E**1**. The die center C**1** of the die **10** can be used for the subsequent bonding process.

[0021] Referring to FIG. **3** to FIG. **5**, in addition, the processor **130** can further determine whether the die **10** beneath the camera device **120** is a good die or a bad die in accordance with an existence of a defect d**1** on the detecting spots P**1**, P**2**, P**3**, and P**4**, such that the die **10** can be selectively picked for the subsequent bonding process. In detail, the processor **130** determine whether there is any defect on the detecting spots P**1**, P**2**, P**3**, and P**4** of the die **10**. If there is no defect detected on the detecting spots P**1**, P**2**, P**3**, and P**4**, the die **10** is determined to be a good die, so the die **10** can be placed onto a bonding apparatus for the subsequent bonding process. Then, as shown in FIG. **5**, the die **10** can be bonded to a substrate **20** according to the die center determined earlier. In other embodiment, the determination of the die center C**1** may be performed after the die **10** is determined to be a good die. If there is a defect detected on at least one of the detecting spots P**1**, P**2**, P**3**, and P**4**, the die **10** is determined to be a bad die, so the die **10** would be skipped (not used) for the subsequent bonding process.

[0022] Referring to FIG. **2** and FIG. **3**, in some embodiments, the camera device **120** includes a light source **122** for emitting light toward the die **10**. The light source **122** may include an illumination source optical element such as a lens, a spectral filter, a polarizer, a diffuser, a spatial filter, a liquid crystal display, a switchable film, polymer dispersed liquid crystals, an electrochromic device, a photochromic device, a sub-combination thereof, a combination thereof, etc. The determination of existence of a defect on the detecting spots may be described as follows. In the present embodiment, the camera device **120** may be sequentially moved to the detecting spots P**1**, P**2**, P**3**, and P**4** to respectively determine whether there is a defect on the detecting spots P**1**, P**2**, P**3**, and P**4**. For example, the camera device

120 may firstly be moved to the detecting spot P**1**, so that the light source **122** emits light toward the detecting spot P**1**. Then, an optical sensor of the camera device **120** is configured to detect an intensity of the light reflected by the detecting spot P**1**. In detail, the light from the light source **122** may impact the detecting spot P**1** and result in regular reflections and/or diffuse reflections. The regular reflections and/or the diffuse reflections may be dependent upon, for example, any target defects. For example, a planar surface without any defect thereon may directly reflect the light and an uneven surface with defect thereon may diffuse the light.

[0023] FIG. **8** is a diagram illustrating a relationship between wavelength and intensity of light according to some exemplary embodiments of the present disclosure. Referring to FIG. **3** and FIG. **8**, in the present embodiment, there is a defect d**1** on the detecting spot P**1** as shown in FIG. **3**, so more diffuse reflection may occur when the light impact the defect d**1** (e.g., tape residue) on the detecting spot P**1**. The optical sensor of the camera device **120** is configured to detect an intensity of the light reflected by the detecting spot P**1**. Accordingly, the processor may determine the existence of the detect d**1** by, for example, distinguishing intensity of reflection of the light associated with the detect d**1** from intensity of reflection of the light associated with the detecting spots P**2**, P**3**, P**4** that does not include a defect d**1**. For example, referring to FIG. **8**, at a specific wavelength, the intensity of the light at the detecting spots P**2**, P**3**, P**4** is much higher than the intensity of the light at the detecting spot P**1** due to the existence of the defect d**1** on the detecting spot P**1**. Accordingly, when the intensity of the light is less than a predetermined value, the processor **130** can determine the existence of the defect d**1** on the detecting spot P**1**. When the processor **130** determines there is a defect d**1** on any of the detecting spots P**1**, P**2**, P**3**, P**4**, the die **10** is determined to be a bad die. When the processor **130** determines there is no defect on any of the detecting spots P**1**, P**2**, P**3**, P**4**, the die **10** is determined to be a good die, and can be continued on the subsequent bonding process.

[0024] Referring to FIG. **6** and FIG. **7**, in some embodiments, the dies **10** that are determined to be good dies may then be bonded to the substrate **20** one by one. During the bonding process, the pick and place tool or subsequent multi-die holder may use a thermal compression bonding head for pick and place the die **10** and then apply heat and pressure for bonding. The die **10** with array of solder bumps **32** may go through a solder reflow furnace to electrically connect the die **10** to the substrate **20**. In an embodiment, the substrate **20** is a device wafer including semiconductor substrate **38** (shown in FIG. **7**), and active devices such as transistors (not shown) formed at the surface of semiconductor substrate **38**. The semiconductor substrate **38** may be a silicon substrate, or may be formed of other semiconductor materials. The device wafer **20** includes a plurality of identical bottom chips **24** therein. A plurality of scribe lines **26** separate the bottom chips **24** from each other. The dies **10** are bonded to the device wafer **20**. In some embodiments, a plurality of dummy dies **10*a*** may also be bonded to the device wafer **20**. Throughout the description, the die **10** may be referred to "active die" or "active top die", which is the die or chip that has electrical functions that contribute to the electrical operation of the resulting package, while the term "dummy die" refers to the die or chip that does not have any electrical function, and the dummy die does not contribute to the electrical operation of the resulting package. The dies **10** may be device dies, or may be packages that including device dies bonded to other package components such as a package substrate, an interposer, and the like.

[0025] FIG. **7** illustrates a cross-sectional view of the structure shown in FIG. **6**, wherein the cross-sectional view is taken along a plane crossing line A-A' in FIG. **6**. The bottom chips **24** may be active chips that include active devices, contact plugs, metal lines, vias and the like (which are formed over top surface of the substrate **38**), although they are not shown in FIG. **7**. As shown in FIG. **7**, the bottom chip **24** may be bonded with one or multiple dies **10** and/or one or multiple dummy dies **10*a***. A plurality of electrical connectors **32** bond the dies **10** and/or **10*a*** to the bottom die **24**. The electrical connectors **32** may be solder bumps, metal-to-metal bonds, solder bumps bonded to metal posts, or the like, and may be used for conducting electrical signals between the bottom chip **24** and the dies **10**. Accordingly, the integrated circuit devices such as transistors (not shown) in the dies **10** are electrically coupled to the devices in the bottom chip **24**. In an embodiment, the electrical connectors **32** are electrically coupled to through-substrate vias (TSVs, also sometimes known as through-silicon vias) **36** in semiconductor substrate **38**. TSVs **36** extend from top surface of substrate **38** to an intermediate level between the top surface and the bottom surface of the substrate **38**.

[0026] A backside interconnect structure of the bottom chip **24** (or bottom wafer **20**) is formed. In an exemplary embodiment, the backside interconnect structure includes dielectric layer(s) **52**, and bond pads/redistribution lines **54**. Bond pads/redistribution lines **54** are electrically coupled to TSVs **36**. Next, connectors **56** may be placed or formed on bond pads **54**. In some embodiments, connectors **56** are solder balls or solder bumps, and are placed/formed and reflowed. In alternative embodiments, connectors **56** may include metal-to-metal bonds, solder bumps bonded to metal posts, or the like, and may be used for conducting electrical signals between bottom chip **24** and another package component such as a device die, an interposer, a package substrate, or a printed circuit board (PCB).

[0027] In one embodiment, the dummy die **10*a*** is attached to the bottom die **24**, for example, through die-attach film **40**, which may be a polymer-based glue. Alternatively, the die-attach film **40** may be a thermal-plastic film that can be cured when heated. In some embodiments, the die-attach film **40** may be an adhesive that loses it adhesion when exposed under light. The dummy die **10*a*** may be a blank die, for example, a semiconductor die (such as a silicon die) that does not have active integrated circuit devices such as transistors and/or passive devices such as resistors, capacitors, and/or the like, formed therein. The dummy die **10*a*** may also be free from low-k dielectric layers, metal lines, vias, and/or the like, therein. In other embodiments, the dummy die **10*a*** may be a dielectric die. In yet other embodiments, the dummy die **10** may reuse bad dies that failed in the previous tests (e.g., having defects on its backside surface), and hence may include integrated circuit devices such as transistors therein. However, the integrated circuit devices in the dummy die **10*a***, if any, do not perform any electrical function in the operation of the resulting semiconductor package **70**, and are not supplied with power. The die-attach film **40** may be formed of an electrical insulating material, and may electrically insulate the dummy die **10*a*** from the bottom chip **24**. Although FIG. **7** illustrates a single dummy die **10*a*** bonded to one the bottom chip **24**, in alternative embodiments, a plurality of dummy dies **10***a* may be attached to the same bottom chip **24**.

[0028] Referring back to FIG. **6**, some of dummy dies **10***a* are attached to the bottom chips **24**, which the bottom chips **24** are complete chips having a rectangular shape, and are functional chips. Additional dummy dies **10***a* may also be attached to incomplete bottom chips **24'**. The incomplete bottom chips **24'** are located at the edges of wafer **20**, and do not have rectangular shapes. The incomplete bottom chips **24'** will not be packaged as products for use. Accordingly, no active dies **10** are bonded to incomplete bottom chips **24'**. The dummy dies **10***a* are configured to reduce issues of stress concentration and/or CTE mismatch of the semiconductor package **70**.

[0029] FIG. **9** and FIG. **10** illustrate schematic views of camera devices capturing images of a plurality of detecting spots on a die according to some exemplary embodiments of the present disclosure. It is noted that the optical inspection system shown in FIG. **9** and FIG. **10** contains many features same as or similar to the optical inspection system disclosed in the previous embodiments. For purpose of clarity and simplicity, detail description of same or similar features may be omitted, and the same or similar reference numbers denote the same or like components.

[0030] Referring to FIG. **9** and FIG. **10**, in some embodiments, the optical inspection system may include a plurality of camera devices **120***a*, **120***b* disposed on a moving mechanism for moving above the die **10**. It is noted that two camera devices **120***a*, **120***b* are illustrated herein, but less or more camera devices may be applied. In the embodiment, the camera devices **120***a*, **120***b* may be disposed on a cantilever arm, which is extended along a first direction parallel to a first edge E**1** of the die **10**, and the cantilever arm may be coupled to a sliding rail, which is extended along a second direction parallel to a second edge E**2** of the die, and the first direction and the second direction intersected with each other. For example, at initial position, the camera devices **120***a*, **120***b* may be arranged along the first edge E**1** of the die **10** to capture images of the detecting spots P**1** and P**2**, and then the camera devices **120***a*, **120***b* are moved along the second edge E**2** of the die to capture images of the detecting spots P**3** and P**4**. With such configuration, the camera devices **120***a*, **120***b* can be moved in one dimension (e.g., linear movement along the edge E**2**) and capture images of the detecting spots distributed in two dimension (e.g., backside surface S**2**).

[0031] For example, the camera devices **120***a*, **120***b* may firstly be moved to the detecting spots P**1**, P**2** respectively, so that the light sources of the camera devices **120***a*, **120***b* emits light toward the detecting spots P**1**, P**2** respectively. Then, the optical sensors of the camera devices **120***a*, **120***b* are configured to detect intensity of the light reflected by the detecting spots P**1**, P**2** respectively. In the present embodiment, there is a defect d**1** on the detecting spot P**1** as shown in FIG. **10**, so more diffuse reflection may occur when the light impact the defect d**1** (e.g., tape residue) on the detecting spot P**1**. The optical sensors of the camera devices **120***a*, **120***b* detect intensity of the light reflected by the detecting spots P**1** and P**2**. Then, the camera devices **120***a*, **120***b* are moved to the detecting spots P**3**, P**4** to perform the same detection process. Accordingly, the processor can determine the intensity of the light at the detecting spots P**3**, P**4** is much higher than the intensity of the light at the detecting spots P**1**, P**2** due to the existence of the defect d**1** on the detecting spot P**1**. Accordingly, when the intensity of the light is less than a predetermined value, the processor **130** can determine the existence of the defect on the detecting spots P**1** and/or P**2**, and the die **10** is determined to be a bad die, and would be skipped for the subsequent bonding process.

[0032] FIG. **11** and FIG. **12** illustrate schematic views of camera devices capturing images of a plurality of detecting spots on a die according to some exemplary embodiments of the present disclosure. It is noted that the optical inspection system shown in FIG. **11** and FIG. **12** contains many features same as or similar to the optical inspection system disclosed in the previous embodiments. For purpose of clarity and simplicity, detail description of same or similar features may be omitted, and the same or similar reference numbers denote the same or like components.

[0033] Referring to FIG. **11** and FIG. **12**, in some embodiments, the optical inspection system may include a plurality of camera devices **120***a*, **120***b* disposed on a moving mechanism for moving above the die **10**. It is noted that two camera devices **120***a*, **120***b* are illustrated herein, but less or more camera devices may be applied. In the embodiment, moving mechanism **140** includes a sliding rail **144** and a cantilever arm **142** coupled to the sliding rail **144**. The camera devices **120***a*, **120***b* are disposed on the cantilever arm **142**, which is extended along a first direction parallel to a first edge E**1** of the die **10**, and the sliding rail **144** is extended along a second direction parallel to a second edge E**2** of the die. The first direction (e.g., edge E**1**) and the second direction (e.g., edge E**2**) are intersected with each other.

[0034] In the present embodiment, for example, each of the camera devices **120***a*, **120***b* is configured to be moved along the cantilever arm **142** to adjust a gap G**1** between adjacent two of the camera devices **120***a*, **120***b*, so as to adjust the positions of the camera devices **120***a*, **120***b* for capturing images of the detecting spots P**1**, P**2** respectively, as shown in FIG. **11**. In addition, referring to FIG. **12**, each of the camera devices **120***a*, **120***b* is configured to rotate relatively to the cantilever arm **142** to adjust an angle θ**1** included between adjacent two of the camera devices **120***a*, **120***b*, so that the camera devices **120***a*, **120***b* is able to capture images of the detecting spots P**3**, P**4** without shifting the cantilever arm **142**.

[0035] In detail, the camera devices **120***a*, **120***b* may firstly be moved by the moving mechanism **140** to the location shown in FIG. **11**, and the camera devices **120***a*, **120***b* can be moved along the cantilever arm **142** to adjust the gap G**1** between the camera devices **120***a*, **120***b* for capturing images of the detecting spots P**1**, P**2** respectively. The light sources of the camera devices **120***a*, **120***b* emits light toward the detecting spots P**1**, P**2**. Then, the optical sensors of the camera devices **120***a*, **120***b* are configured to detect intensity of the light reflected by the detecting spots P**1**, P**2** respectively. The optical sensors of the camera devices **120***a*, **120***b* detect intensity of the light reflected by the detecting spots P**1** and P**2**. Then, the camera devices **120***a*, **120***b* are rotated relatively to the cantilever arm **142** to adjust the angle **01** included between the camera devices **120***a*, **120***b* to perform the same detection process on the detecting spots P**3** and P**4**. In the present embodiment, there are defects d**1**, d**2**, d**3**, d**4** on the detecting spots P**1**, P**2**, P**3**, and P**4** as shown in FIG. **11**. Accordingly, when the intensity of the light is less than a predetermined value, the processor determines the existence of the defect on the detecting spots P**1**, P**2**, P**3**, and/or

P4, so the die **10** is determined to be a bad die, and would be skipped for the subsequent bonding process.

[0036] FIG. **13** illustrates a schematic view of a camera device capturing images of a plurality of detecting spots on a die according to some exemplary embodiments of the present disclosure. FIG. **14** illustrates a schematic view of a camera device disposed on a moving mechanism according to some exemplary embodiments of the present disclosure. It is noted that the optical inspection system shown in FIG. **13** and FIG. **14** contains many features same as or similar to the optical inspection system disclosed in the previous embodiments. For purpose of clarity and simplicity, detail description of same or similar features may be omitted, and the same or similar reference numbers denote the same or like components.

[0037] Referring to FIG. **13** and FIG. **14**, in some embodiments, the camera device **120** is disposed on a moving mechanism **140** for moving above the die **10**. In the embodiment, the camera device **120** may be coupled to a sliding rail **144**, which is extended along a direction parallel to an edge E1 of the die. Accordingly, the camera device **120** can slide along the edge E1 of the die **10**. In the present embodiment, the optical inspection system further includes an optical sensor **160** and a reflector **150**. The optical sensor **160** and the camera device **120** are disposed on two opposite sides of the die **10** and the optical sensor **160** may be coupled to another sliding rail, which is parallel to the sliding rail **144**, to slide along the opposite edge of the die **10**. The reflector **150** may be a reflecting mirror and is disposed between the camera device **120** and the optical sensor **160**. In the present embodiment, the optical inspection system may include a plurality of reflectors **150** arranged along the direction parallel to an edge E1 of the die. In an alternative embodiment, single reflector **150** may be coupled to a sliding rail to slide along with the camera device **120** and optical sensor **160**. Accordingly, the camera device **120** includes a light source **122** for emitting light toward the die **10**, the light is reflected sequentially by a first one of the detecting spots (e.g., detecting spot P**1**), the reflector **150**, and a second one of the detecting spots (e.g., detecting spot P**2**), and is then detected by the optical sensor **160**.

[0038] In the present embodiment, the camera device **120** is disposed at an angle θ**2** with respect to the detecting spots (e.g., detecting spot P**1**). Accordingly, the light source **122** of the camera device **120** provides an indirect illumination source that may be configured to at least partially illuminate the detecting spots at the angle θ**2** with respect to the detecting spots (e.g., detecting spot P**1**). The light from the light source **122** impacts the detecting spot P**1** and result in direct reflections and/or diffuse reflection. As illustrated in FIG. **13**, the optical sensor **160** and the reflector **150** may be geometrically oriented with respect to the light source **122** of the camera device **120**, such that the reflector **150** can reflect the light reflected by one of the detecting spots and redirect the light to impact the next one of the detecting spots, and the optical sensor **160** can detect intensity of the light reflected by the next one of the detecting spots. For example, the light from the light source **122** is emitted toward the detecting spot P**1**, and the reflector **150** is configured to reflect the light that is reflected by the detecting spot P**1** and redirected the light toward the detecting spot P**2**. The optical sensor **160** is configured to detect intensity of the light reflected by the detecting spot P**2**. Thereby, the system is capable of detecting multiple detecting spots in one direction without moving the camera device **120**, so as to improve inspection efficiency.

[0039] In the present embodiment, there are defects d**1**, d**2** at the detecting spots P**1** and P**2**, so more diffuse reflection may occur when the light impact the defects d**1**, and d**2** on the detecting spots P**1** and P**2**. Accordingly, the intensity of the light at the detecting spots P**1** and P**2** detected by the optical sensor **160** is less than a predetermined value, so the processor **130** can determine the existence of the defects on the detecting spots P**1** and/or P**2**, and the die **10** is determined to be a bad die, and would be skipped for the subsequent bonding process.

[0040] FIG. **15** illustrates a schematic view of a camera device capturing images of a plurality of detecting spots on a die according to some exemplary embodiments of the present disclosure. It is noted that the optical inspection system shown in FIG. **15** contains many features same as or similar to the optical inspection system disclosed in the previous embodiments. For purpose of clarity and simplicity, detail description of same or similar features may be omitted, and the same or similar reference numbers denote the same or like components.

[0041] Referring to FIG. **15**, in some embodiments, the camera device **120** may be coupled to the sliding rail for moving along a direction parallel to the edge E**1** of the die. In the present embodiment, the optical inspection system further includes an optical sensor **160** and a reflector **150**. The optical sensor **160** and the camera device **120** are disposed on two opposite sides of the die **10** and the optical sensor **160** is also coupled to a sliding rail to slide along the opposite edge of the die **10**. The reflector **150** is disposed between the camera device **120** and the optical sensor **160**. In the present embodiment, the optical inspection system may include a plurality of line reflectors **150*a*** parallel to one another, and the long axis A**1** of each of the line reflectors **150*a*** is perpendicular to the edge E**1** of the die **10**. Accordingly, the camera device **120** includes the light source **122** for emitting light toward the die **10**, the light is reflected alternately between the detecting spots P**1**, P**2** . . . , PN and the reflector, and is then detected by the optical sensor **160**.

[0042] In the present embodiment, the camera device **120** is disposed at an acute angle) (<90°) with respect to the detecting spots (e.g., detecting spot P**1**). The light from the light source **122** impacts the detecting spot P**1** and result in direct reflections and/or diffuse reflection. As illustrated in FIG. **15**, the optical sensor **160** and the reflector **150*a*** may be geometrically oriented with respect to the light source **122** of the camera device **120**, such that the reflector **150*a*** can reflect the light reflected by one of the detecting spots and redirect the light to impact the adjacent one of the detecting spots, and the optical sensor **160** can detect intensity of the light reflected by the last one of the detecting spots in the corresponding row. For example, the light from the light source **122** is emitted toward the detecting spot P**1**, and the reflector **150*a*** is configured to reflect the light that is reflected by the detecting spot P**1** and redirected the light toward the detecting spot P**2**, and the light would be reflected back and forth between the reflector **150*a*** and the detecting spots of the die **10** until the light is received by the optical sensor **160**. The optical sensor **160** detects intensity of the light reflected by the detecting spot PN. The camera device **120** can be moved to the next location (illustrated in dotted line) to detecting multiple detecting spots in the next row. Thereby, the system is capable of detecting multiple detecting spots P**1**, P**2** . . . , PN in one row without moving the camera device **120**, so as to improve inspection efficiency.

[0043] In the present embodiment, there are defects on at least the detecting spots P**1**, P**2**, and PN so more diffuse reflection may occur when the light impact the defects on those detecting spots. Accordingly, the intensity of the light at those detecting spots detected by the optical sensor **160** is less than a predetermined value, so the processor **130** can determine the existence of the defects on the detecting spots P**1**, P**2** . . . , and/or PN, and the die **10** is determined to be a bad die, and would be skipped for the subsequent bonding process.

[0044] FIG. **16** illustrates a schematic view of a line-scan camera device capturing images of a plurality of detecting spots on a die according to some exemplary embodiments of the present disclosure. It is noted that the optical inspection system shown in FIG. **16** contains many features same as or similar to the optical inspection system disclosed in the previous embodiments. For purpose of clarity and simplicity, detail description of same or similar features may be omitted, and the same or similar reference numbers denote the same or like components.

[0045] Referring to FIG. **16**, in some embodiments, the camera device may be a line-scan camera device **120***c* extended along an edge E**1** of the die **10**, the optical sensor **160***c* includes an array of optical sensors **160***c* arranged along a direction parallel to the edge E**1** of the die **10**, and the reflector **150***c* may be a plane reflector. In the embodiment, the array of optical sensors **160***c* may be integrated as a line-scan optical sensor. The plane reflector **150***c* is disposed between the line-scan camera device **120***c* and the array of optical sensors **160***c*. Accordingly, the line-scan camera device **120***c* includes the light source **122** for emitting light toward the die **10**, the light is reflected alternately between the detecting spots P**1**, P**2** . . . , PN and the plane reflector **150***c*, and is then detected by the array of the optical sensors **160***c*.

[0046] In the present embodiment, the line-scan camera device **120***c* is disposed at an acute angle (<90°) with respect to the detecting spots P**1**, P**2** . . . , PN. As illustrated in FIG. **16**, the array of the optical sensors **160***c* and the plane reflector **150***c* may be geometrically oriented with respect to the light source **122** of the line-scan camera device **120***c*, such that the plane reflector **150***c* can reflect the light reflected by one of the detecting spots and redirect the light to impact the adjacent one of the detecting spots, and the array of the optical sensors **160***c* can detect intensity of the light reflected by the last one of the detecting spots in the corresponding row. For example, the light from the light source **122** is emitted toward the detecting spot P**1**, and the plane reflector **150***c* is configured to reflect the light that is reflected by the detecting spot P**1** and redirected the light toward the detecting spot P**2**, and the light would be reflected back and forth between the plane reflector **150***c* and the detecting spots of the die **10** until the light is received by the array of the optical sensors **160***c*. The array of the optical sensors **160** detect intensity of the light reflected by the detecting spot PN. Thereby, the system is capable of detecting multiple detecting spots P**1**, P**2** . . . , PN distributed over the backside surface S**2** all at once without moving the line-scan camera device **120***c*, so as to improve inspection efficiency.

[0047] In the present embodiment, there are defects on at least the detecting spots P**1**, P**2**, and PN so more diffuse reflection may occur when the light impact the defects on those detecting spots. Accordingly, the intensity of the light at those detecting spots detected by the array of the optical sensors **160***c* is less than a predetermined value, so the processor can determine the existence of the defects on the detecting spots P**1**, P**2** . . . , and/or PN, and the die **10** is determined to be a bad die, and would be skipped for the subsequent bonding process.

[0048] FIG. **17** illustrates a schematic view of a camera device disposed on a moving mechanism according to some exemplary embodiments of the present disclosure. FIG. **18** and FIG. **19** illustrate schematic views of a camera device disposed on a moving mechanism according to some exemplary embodiments of the present disclosure. It is noted that the optical inspection system shown in FIG. **17** to FIG. **19** contains many features same as or similar to the optical inspection system disclosed in the previous embodiments. For purpose of clarity and simplicity, detail description of same or similar features may be omitted, and the same or similar reference numbers denote the same or like components.

[0049] Referring to FIG. **17** to FIG. **19**, in some embodiments, the camera device **120** is movably disposed on a ring sliding rail **170** at a side of the die **10** for changing a light emitting direction (angle) of the light emitted by the light source **122**. In the present embodiment, the ring sliding rail **170** may further include a cantilever arm **172** extended toward the die **10**. The optical sensor **160** and the camera device **120** are disposed on two opposite sides of the die **10** and the optical sensor **160** may be coupled to a linear sliding rail to slide along an edge of the die **10**. That is, the optical sensor **160** and the camera device **120** are disposed on two opposite sides of the die **10**, but the camera device **120** is configured to rotate relatively to the die **10** while the optical sensor **160** is configured to slide along the edge of the die **10**. In one embodiment, the ring sliding rail **170** may further be moved along the edge of the die **10** by, for example, coupled to a linear sliding rail extended along the edge of the die **10**. The reflectors **150** are disposed between the camera device **120** and the optical sensor **160**, and are geometrically oriented with respect to the light source **122** of the camera device **120**, such that the reflector **150** can reflect the light reflected by the detecting spots and redirect the light to impact the adjacent one of the detecting spots, and the optical sensor **160** can detect intensity of the light reflected by the last one of the detecting spots in the corresponding row.

[0050] In the present embodiment, when the camera device **120** is located at the first position where the camera device **120** is illustrated in solid line in FIG. **17** and also shown in FIG. **18**, the light from the light source **122** is emitted downward at an acute angle) (<90°) with respect to the detecting spot P**1**. The light from the light source **122** impacts the detecting spot P**1** and result in direct reflections and/or diffuse reflection. The reflector **150** can reflect the light reflected by the detecting spot P**1** and redirect the light to impact the adjacent one of the detecting spots (e.g., the detecting spot P**2**), and the optical sensor **160** can detect intensity of the light reflected by the detecting spot P**2**. Then, the camera may be rotated along the ring sliding rail **170** to a second position where the camera device **120** is illustrated in dotted line in FIG. **17** and also shown in FIG. **19**, the light from the light source **122** is emitted upward at an acute angle (<90°) with respect to the reflector **150**. The light from the light source **122** impacts the reflector **150** and is reflected and redirected toward the detecting spot P**3**, and may be reflected and redirected again by the next reflector to impact the detecting spots P**4**, so the optical sensor **160** can detect intensity of the light reflected by the detecting spot P**4**. Thereby, the location of the detecting spots and the gap between the detecting spots may be adjusted by rotating the camera device **120** to change the incident angle of the light, so as to improve inspection flexibility.

[0051] In the present embodiment, there are defects on the detecting spots P**1**, P**2**, P**3**, and P**4**, so more diffuse reflection may occur when the light impact the defects on those detecting spots. Accordingly, the intensity of the light at those detecting spots detected by the optical sensor **160** is less than a predetermined value, so the processor can determine the existence of the defects on the detecting spots P**1**, P**2**, P**3**, and/or P**4**, and the die **10** is determined to be a bad die, and would be skipped for the subsequent bonding process.

[0052] Based on the above discussions, it can be seen that the present disclosure offers various advantages. It is understood, however, that not all advantages are necessarily discussed herein, and other embodiments may offer different advantages, and that no particular advantage is required for all embodiments.

[0053] In accordance with some embodiments of the disclosure, an optical inspection system includes a die carrier, a camera device, and a processor. The die carrier is configured to carry a die, wherein a backside surface of the die faces away from the die carrier. The camera device is movably disposed above the die carrier and configured to capture images of a plurality of detecting spots on the backside surface, wherein the plurality of detecting spots comprising a plurality of die corners of the die. The processor is coupled to the camera device and configured to determine a die center of the die according to the images of the plurality of die corners of the die, and determine whether the die is a bad die according to an existence of a defect on the backside surface of the die detected by the camera device. In one embodiment, the camera device comprises a plurality of camera devices arranged along a cantilever arm, which is configured to be moved along a sliding rail. In one embodiment, the cantilever arm is extended along a first direction parallel to a first edge of the die and the sliding rail is extended along a second direction parallel to a second edge of the die, and the first direction and the second direction intersected with each other. In one embodiment, each of the plurality of camera devices is configured to be moved along the cantilever arm to adjust a gap between adjacent two of the plurality of camera devices and rotated relatively to the cantilever arm to adjust an angle included between adjacent two of the plurality of camera devices. In one embodiment, the optical inspection system further includes an optical sensor and a reflector disposed between the camera device and the optical sensor, wherein the camera device comprises a light source for emitting light toward the die, the light is reflected sequentially by a first one of the plurality of detecting spots, the reflector, and a second one of the plurality of detecting spots, and is then detected by the optical sensor. In one embodiment, the reflector comprises a plurality of reflectors arranged along a direction parallel to an edge of the die. In one embodiment, the reflector comprises a plurality of line reflectors parallel to one another, and a long axis of each of the plurality of line reflectors is perpendicular to an edge of the die. In one embodiment, the processor determines the existence of a defect when an intensity of the light detected by the optical sensor is less than a predetermined value. In one embodiment, the camera device is coupled to a first sliding rail extended along an edge of the die and the optical sensor is coupled to a second sliding rail parallel to the first sliding rail. In one embodiment, the camera device comprises a line-scan camera device extended along an edge of the die, the optical sensor comprises an array of optical sensor arranged along a direction parallel to the edge of the die, and the reflector comprises a plane reflector. In one embodiment, the camera device is movably disposed on a ring sliding rail at a side of the die for changing a light emitting direction of the light emitted by the light source. In one embodiment, the ring sliding rail is configured to be moved along an edge of the die and the optical sensor is configured to be moved along a direction parallel to the edge of the die.

[0054] In accordance with some embodiments of the disclosure, a method of optical inspection includes the following steps: picking up a die from a tape carrier; flipping the die so that a backside surface of the die faces a camera device; capturing images of a plurality of detecting spots on the backside surface of the die by the camera device, wherein the plurality of detecting spots comprising a plurality of die corners of the die; and determining whether the die is a bad die or not according to an existence of a defect on the plurality of detecting spots of the die; and determining a die center of the die according to the images of the plurality of die corners of the die. In one embodiment, the defect comprises tape residue left on the backside surface of the die. In one embodiment, picking up the die from the tape carrier comprises ejecting a plurality of ejector pins underneath the tape carrier for pushing the die away from the tape carrier, and the plurality of detecting spots corresponding to spots where the plurality of ejector pins pushes the die. In one embodiment, determining whether the die is bad die or not according to the existence of the defect on the plurality of detecting spots of the die comprises: emitting light toward one of the plurality of detecting spots; detecting an intensity of the light reflected by the one of the plurality of detecting spots; determining the defect existing on the one of the plurality of detecting spots when the intensity of the light is less than a predetermined value; and determining the die is bad die when the defect existing on the one of the plurality of detecting spots is determined. In one embodiment, determining whether the die is bad die or not according to the existence of the defect on the plurality of detecting spots of the die comprises: emitting light toward a first one of the plurality of detecting spots, wherein the light reflected by the first one of the plurality of detecting spots is reflected and redirected toward a second one of the plurality of detecting spots by a reflector; detecting an intensity of the light reflected by the second one of the plurality of detecting spots; determining the defect existing on the first one or the second one of the plurality of detecting spots when the intensity of the light is less than a predetermined value; and determining the die is bad die when the defect existing on the first one or the second one of the plurality of detecting spots is determined.

[0055] In accordance with some embodiments of the disclosure, a method of manufacturing a semiconductor package includes the following steps: performing a dicing process to form a plurality of dies on a tape carrier; picking up one of the plurality of dies from the tape carrier; flipping the die so that a backside surface of the die faces a camera device; capturing images of a plurality of detecting spots on the backside surface of the die by the camera device; and determining the die is good die when there is no defect detected on the plurality of detecting spots of the die; determining a die center of the die according to the images of the plurality of detecting spots; and performing a bonding process for bonding the die to a substrate according to the die center. In one embodiment, the plurality of detecting spots comprising a plurality of die corners of the die, and the die center of the die is determined according to the images of the plurality of die corners of the die. In one embodiment, the method further comprising determining the die is bad die when there is a defect detected on one of the plurality of detecting spots of the die, and skipping the die for the bonding process.

[0056] The foregoing outlines features of several embodiments so that those skilled in the art may better understand the aspects of the present disclosure. Those skilled in the art should appreciate that they may readily use the present disclosure as a basis for designing or modifying other processes and structures for carrying out the same purposes and/or achieving the same advantages of the embodiments introduced herein. Those skilled in the art should also realize that such equivalent constructions do not depart from the spirit and scope of the present disclosure, and that they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the present disclosure.

What is claimed is:

1. An optical inspection system, comprising:

a die carrier configured to carry a die, wherein a backside surface of the die faces away from the die carrier;

a camera device movably disposed above the die carrier and configured to capture images of a plurality of detecting spots on the backside surface, wherein the plurality of detecting spots comprising a plurality of die corners of the die; and

a processor coupled to the camera device and configured to determine a die center of the die according to the images of the plurality of die corners of the die, and determine whether the die is a bad die according to an existence of a defect on the backside surface of the die detected by the camera device.

2. The optical inspection system as claimed in claim **1**, wherein the camera device comprises a plurality of camera devices arranged along a cantilever arm, which is configured to be moved along a sliding rail.

3. The optical inspection system as claimed in claim **2**, wherein the cantilever arm is extended along a first direction parallel to a first edge of the die and the sliding rail is extended along a second direction parallel to a second edge of the die, and the first direction and the second direction intersected with each other.

4. The optical inspection system as claimed in claim **2**, wherein each of the plurality of camera devices is configured to be moved along the cantilever arm to adjust a gap between adjacent two of the plurality of camera devices and rotated relatively to the cantilever arm to adjust an angle included between adjacent two of the plurality of camera devices.

5. The optical inspection system as claimed in claim **1**, further comprises an optical sensor and a reflector disposed between the camera device and the optical sensor, wherein the camera device comprises a light source for emitting light toward the die, the light is reflected sequentially by a first one of the plurality of detecting spots, the reflector, and a second one of the plurality of detecting spots, and is then detected by the optical sensor.

6. The optical inspection system as claimed in claim **5**, wherein the reflector comprises a plurality of reflectors arranged along a direction parallel to an edge of the die.

7. The optical inspection system as claimed in claim **5**, wherein the reflector comprises a plurality of line reflectors parallel to one another, and a long axis of each of the plurality of line reflectors is perpendicular to an edge of the die.

8. The optical inspection system as claimed in claim **5**, wherein the processor determines the existence of a defect when an intensity of the light detected by the optical sensor is less than a predetermined value.

9. The optical inspection system as claimed in claim **5**, wherein the camera device is coupled to a first sliding rail extended along an edge of the die and the optical sensor is coupled to a second sliding rail parallel to the first sliding rail.

10. The optical inspection system as claimed in claim **5**, wherein the camera device comprises a line-scan camera device extended along an edge of the die, the optical sensor comprises an array of optical sensor arranged along a direction parallel to the edge of the die, and the reflector comprises a plane reflector.

11. The optical inspection system as claimed in claim **5**, wherein the camera device is movably disposed on a ring sliding rail at a side of the die for changing a light emitting direction of the light emitted by the light source.

12. The optical inspection system as claimed in claim **11**, wherein the ring sliding rail is configured to be moved along an edge of the die and the optical sensor is configured to be moved along a direction parallel to the edge of the die.

13. A method of optical inspection, comprising:

picking up a die from a tape carrier;

flipping the die so that a backside surface of the die faces a camera device;

capturing images of a plurality of detecting spots on the backside surface of the die by the camera device, wherein the plurality of detecting spots comprising a plurality of die corners of the die; and

determining whether the die is a bad die or not according to an existence of a defect on the plurality of detecting spots of the die; and

determining a die center of the die according to the images of the plurality of die corners of the die.

14. The method as claimed in claim **13**, wherein the defect comprises tape residue left on the backside surface of the die.

15. The method as claimed in claim **13**, wherein picking up the die from the tape carrier comprises ejecting a plurality of ejector pins underneath the tape carrier for pushing the die away from the tape carrier, and the plurality of detecting spots corresponding to spots where the plurality of ejector pins pushes the die.

16. The method as claimed in claim **13**, wherein determining whether the die is bad die or not according to the existence of the defect on the plurality of detecting spots of the die comprises:

emitting light toward one of the plurality of detecting spots;

detecting an intensity of the light reflected by the one of the plurality of detecting spots;

determining the defect existing on the one of the plurality of detecting spots when the intensity of the light is less than a predetermined value; and

determining the die is bad die when the defect existing on the one of the plurality of detecting spots is determined.

17. The method as claimed in claim **13**, wherein determining whether the die is bad die or not according to the existence of the defect on the plurality of detecting spots of the die comprises:

emitting light toward a first one of the plurality of detecting spots, wherein the light reflected by the first one of the plurality of detecting spots is reflected and redirected toward a second one of the plurality of detecting spots by a reflector;

detecting an intensity of the light reflected by the second one of the plurality of detecting spots;

determining the defect existing on the first one or the second one of the plurality of detecting spots when the intensity of the light is less than a predetermined value; and

determining the die is bad die when the defect existing on the first one or the second one of the plurality of detecting spots is determined.

18. A method of manufacturing a semiconductor package, comprising:

performing a dicing process to form a plurality of dies on a tape carrier;

picking up one of the plurality of dies from the tape carrier;

flipping the die so that a backside surface of the die faces a camera device;

capturing images of a plurality of detecting spots on the backside surface of the die by the camera device; and

determining the die is good die when there is no defect detected on the plurality of detecting spots of the die;

determining a die center of the die according to the images of the plurality of detecting spots; and

performing a bonding process for bonding the die to a substrate according to the die center.

19. The method as claimed in claim **18**, wherein the plurality of detecting spots comprising a plurality of die corners of the die, and the die center of the die is determined according to the images of the plurality of die corners of the die.

20. The method as claimed in claim **18**, further comprising determining the die is bad die when there is a defect detected on one of the plurality of detecting spots of the die, and skipping the die for the bonding process.

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